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Portfolio: romia.dev | case studies, architecture breakdowns, and live demos

Summary

AI Engineer building production multi-agent and Retrieval-Augmented Generation (RAG) systems with full-stack breadth: FastAPI and Spring Boot backends, Kafka event-driven microservices, and React/Next.js. Co-founded Zaylon AI and built its LangGraph tool-calling agent and tri-lingual NLP on a multi-tenant FastAPI backend.

Experience

Software Engineer & Co-Founder

May 2025 – Present

Zaylon AI

- Lead engineering (architecture, reviews, roadmap, DevOps); built the **FastAPI** backend with **Docker** and CI/CD.
- Built the conversational AI as a single **LangGraph** tool-calling agent with around 30 scoped tools.
- Engineered **DialectBridge**: tri-lingual NLP with dialect detection for English, Egyptian Arabic, and Franco-Arabic.
- Hardened the backend: **argon2id** auth, row-level multi-tenant isolation over 80+ tables, a Postgres checkout saga.

Software Integration Engineering Intern

Jul 2025

Ejada

- Built **Spring Boot** REST APIs for core banking with query optimization and **Redis** caching.
- Ran **Kafka** producers/consumers for async messaging; secured the gateway with OAuth2 and rate limiting (WSO2).

AI Coding Expert

Sep 2024 – Feb 2026

Freelance

- Ranked thousands of Python samples and built quality rubrics for RLHF pipelines training frontier AI models.

Automotive Cybersecurity Intern

Aug 2024 – Oct 2024

Swift-ACT

- Built threat-model catalogs and mitigations for **CAN bus** and **ECU** in-vehicle networks.
- Ran attack simulations (spoofing, replay, DoS) in **Dyna4/CARLA**; fault injection and ECU validation via **Vector CANoe**.

Projects

Talos: Multi-User RAG Document Platform

Graduation Project | [Case Study](#)

- Owned the ingestion and retrieval core of a **FastAPI** document-chat platform; cited answers streamed over SSE.
- Hybrid retrieval (dense + **BM25**, RRF) with cross-encoder rerank; async **ARQ/MinIO/Milvus** ingestion pipeline.

Virtual Banking Microservices

[GitHub](#) | [Case Study](#)

- Five **Spring Boot 3** services behind a gateway; **Kafka** transfer saga: outbox, idempotent consumers, RS256 JWT.
- Atomic transfers via pessimistic locking and a balance constraint, with Testcontainers integration tests.

ContextIQ: Hybrid-Retrieval RAG

[GitHub](#) | [Demo](#) | [Case Study](#)

- **FastAPI** RAG on CPU (ONNX): dense + **BM25** fused with RRF, cross-encoder reranking, grounded citations.
- Evaluation harness over a corpus of confusable distractors; abstains when the context is insufficient.

Multimodal Emotion Recognition

[GitHub](#) | [Demo](#) | [Case Study](#)

- Speaker-independent (actor-disjoint): frozen **WavLM-large** + facial-expression model, calibrated late fusion.
- Learnable layer-weighting and attentive pooling over WavLM; a unit test guards against speaker leakage.

VoicePrint: Authorship Stylometry

[Case Study](#)

- **StyleDistance** embedding plus function-word features; scores voice match and rewrites a draft toward you.

Education

Alexandria University, Faculty of Engineering

Expected Jul 2026

B.Sc. in Computer & Communications Engineering (AI Concentration) | Cumulative GPA: **3.73 / 4.0**

Technical Skills

Languages: Python, Java, JavaScript/TypeScript, SQL, C/C++

Backend: FastAPI, Spring Boot, Microservices, Event-Driven Architecture, Kafka, REST APIs, PostgreSQL, Redis

AI / ML: LangGraph, LangChain, PyTorch, RAG, Multi-Agent Systems, NLP, Vector DBs (ChromaDB, pgvector), Reranking

Frontend & DevOps: React, Next.js, Tailwind CSS, Docker, Git, GitHub Actions, CI/CD, Linux

Certifications: micro1 Certified Software Engineer | [Google AI Agents](#) | Meta Back-end | AWS Cloud & ML

Competitions: MTC-AIC3 BCI (Egypt National AI) | ECPC (3x) | Hepsiburada NLP Hackathon (TEKNOFEST)

Languages: English (Fluent) | Arabic (Native) | German (A2) | Spanish (Conversational)